F78PA Probability & Statistics A

Tutorial 6 – Solutions

1. First define some notation for events and random variables of interest:

$$\begin{aligned} T &= \text{lifetime of chosen battery} \\ T_A &\sim \text{Exp}(1/20) \\ T_B &\sim \text{Exp}(1/30) \\ A &= \{\text{chosen battery is of type A}\} \\ B &= \{\text{chosen battery is of type B}\} \\ C &= \{\text{still working after 20 hours}\} = \{T > 20\} \\ D &= \{\text{still working after 30 hours}\} = \{T > 30\} \end{aligned}$$

We are given the following information:

$$\mathbb{P}(A) = 2/3 \qquad \mathbb{P}(B) = 1/3$$

The lifetime of a type A battery is exponentially distributed with rate $\lambda_A = 1/20$, while the lifetime of a type B battery is exponentially distributed with rate $\lambda_B = 1/30$.

(a) Asked to find $\mathbb{P}(A \mid T > 20) = \mathbb{P}(A \mid C)$. Now

$$\begin{aligned} \mathbb{P}(A \mid C) &= \frac{\mathbb{P}(A \cap C)}{\mathbb{P}(C)} = \frac{\mathbb{P}(C \mid A)\mathbb{P}(A)}{\mathbb{P}(C)} \\ &= \frac{\mathbb{P}(C \mid A)\mathbb{P}(A)}{\mathbb{P}(C \mid A)\mathbb{P}(A) + \mathbb{P}(C \mid B)\mathbb{P}(B)} \quad \text{(Bayes' Theorem)} \end{aligned}$$

We are told that the lifetime of a type A battery is exponentially distributed with rate $\lambda_A = \frac{1}{20}$, so

$$\begin{aligned} \mathbb{P}(C \mid A) = \mathbb{P}(T > 20 \mid A) &= \mathbb{P}(T_A > 20) \\ &= \int_{x=20}^{\infty} (1/20)\, e^{-x/20}\, dx = \Big[-e^{-x/20}\Big]_{x=20}^{\infty} = e^{-1} \end{aligned}$$

Similarly,

$$\begin{aligned} \mathbb{P}(C \mid B) = \mathbb{P}(T > 20 \mid B) &= \mathbb{P}(T_B > 20) \\ &= \int_{x=20}^{\infty} (1/30)\, e^{-x/30}\, dx = \Big[-e^{-x/30}\Big]_{x=20}^{\infty} = e^{-2/3} \end{aligned}$$

So applying Bayes' Theorem,

$$\begin{aligned} \mathbb{P}(A \mid C) &= \frac{\mathbb{P}(C \mid A)\mathbb{P}(A)}{\mathbb{P}(C \mid A)\mathbb{P}(A) + \mathbb{P}(C \mid B)\mathbb{P}(B)} \\ &= \frac{(2/3)e^{-1}}{(2/3)e^{-1} + (1/3)e^{-2/3}} = \frac{2}{2 + e^{1/3}} \approx 0.5890 \end{aligned}$$

(b) Asked to find $\mathbb{P}(T > 30 \mid T > 20) = \mathbb{P}(D \mid C)$. Now

$$\begin{aligned} \mathbb{P}(D \mid C) = \frac{\mathbb{P}(C \cap D)}{\mathbb{P}(C)} &= \frac{\mathbb{P}(\{T > 20\} \cap \{T > 30\})}{\mathbb{P}(\{T > 20\})} \\ &= \frac{\mathbb{P}(T > 30)}{\mathbb{P}(T > 20)} = \frac{\mathbb{P}(D)}{\mathbb{P}(C)} \end{aligned}$$

noticing that $C \cap D = D$ here.

Applying the Law of Total Probability to numerator and denominator,

$$\mathbb{P}(D \mid C) = \frac{\mathbb{P}(D)}{\mathbb{P}(C)} = \frac{\mathbb{P}(D \mid A)\mathbb{P}(A) + \mathbb{P}(D \mid B)\mathbb{P}(B)}{\mathbb{P}(C \mid A)\mathbb{P}(A) + \mathbb{P}(C \mid B)\mathbb{P}(B)}$$

Similarly to part (a),

$$\begin{aligned}
\mathbb{P}(D \mid A) = \mathbb{P}(T > 30 \mid A) &= \mathbb{P}(T_A > 30) \\
&= \int_{x=30}^{\infty} (1/20)\, e^{-x/20}\, dx = \left[-e^{-x/20}\right]_{x=30}^{\infty} = e^{-3/2} \\
\mathbb{P}(D \mid B) = \mathbb{P}(T > 30 \mid B) &= \mathbb{P}(T_B > 30) \\
&= \int_{x=30}^{\infty} (1/30)\, e^{-x/30}\, dx = \left[-e^{-x/30}\right]_{x=30}^{\infty} = e^{-1}
\end{aligned}$$

and so

$$\begin{aligned}
\mathbb{P}(D \mid C) &= \frac{\mathbb{P}(D \mid A)\mathbb{P}(A) + \mathbb{P}(D \mid B)\mathbb{P}(B)}{\mathbb{P}(C \mid A)\mathbb{P}(A) + \mathbb{P}(C \mid B)\mathbb{P}(B)} \\
&= \frac{(2/3)e^{-3/2} + (1/3)e^{-1}}{(2/3)e^{-1} + (1/3)e^{-2/3}} = \frac{2e^{-1/2} + 1}{2 + e^{1/3}} \approx 0.6517
\end{aligned}$$

2. $X_1, \ldots, X_n$ are independent random variables with distribution functions $F_1, \ldots, F_n$, respectively, and $M = \min\{X_1, \ldots, X_n\}$.

   (a) For $x \in \mathbb{R}$,

$$\begin{aligned}
\mathbb{P}(M > x) &= \mathbb{P}\left(\min\{X_1, \ldots, X_n\} > x\right) \\
&= \mathbb{P}(X_1 > x,\ , X_2 > x, \ldots, X_n > x) \\
&= \mathbb{P}(X_1 > x)\mathbb{P}(X_2 > x) \cdots \mathbb{P}(X_n > x) \text{ by independence of } X_1, \ldots, X_n
\end{aligned}$$

   (b) Denoting by F_M the distribution function of M, then for $x \in \mathbb{R}$,

$$\begin{aligned}
F_M(x) &= \mathbb{P}(M \le x) \\
&= 1 - \mathbb{P}(M > x) \\
&= 1 - \mathbb{P}(X_1 > x)\mathbb{P}(X_2 > x) \cdots \mathbb{P}(X_n > x) \text{ from prt (a)} \\
&= 1 - (1 - \mathbb{P}(X_1 \le x))\,(1 - \mathbb{P}(X_2 \le x) \cdots (1 - \mathbb{P}(X_n \le x)) \\
&= 1 - (1 - F_1(x))\,(1 - F_2(x)) \cdots (1 - F_n(x))
\end{aligned}$$

   (c) For $i = 1, 2, 3, 4, 5$, the time that the ith clock rings is denoted T_i, and we are told that $T_i \sim \text{Exp}(1)$, and that $T_1, \ldots, T_5$ are independent random variables. The time that the first clock goes off is $M = \min\{T_1, \ldots, T_5\}$.
   Since $\text{Im}T_i = [0, \infty)$ for $i = 1, \ldots, 5$, we have $\text{Im}M = [0, \infty)$, and so for $x < 0$, $F_M(x) = 0$.
   For $T_i \sim \text{Exp}(1)$, we have $F_i(x) = \mathbb{P}(T_i \le x) = 1 - e^{-x}$, and so for $x \ge 0$, applying part (b),

$$\begin{aligned}
F_M(x) &= 1 - \left(e^{-x}\right)\left(e^{-x}\right)\left(e^{-x}\right)\left(e^{-x}\right)\left(e^{-x}\right) \\
&= 1 - e^{-5x}
\end{aligned}$$

   This is the distribution function for an Exp(5) random variable, so we have shown that $M \sim \text{Exp}(5)$.
   So $\mathbb{E}(M) = 1/5 = 0.2$ hours $= 12$ minutes.

3. Recall that $F_{X,Y}(x,y) = \mathbb{P}(X \leq x,\, Y \leq y)$.

   So

$$\begin{aligned} F_{X,Y}(b,d) - F_{X,Y}(b,c) &= \mathbb{P}(X \leq b,\, Y \leq d) - \mathbb{P}(X \leq b,\, Y \leq c) \\ &= \mathbb{P}(X \leq b,\, c < Y \leq d) \end{aligned}$$

   Similarly,

$$\begin{aligned} F_{X,Y}(a,d) - F_{X,Y}(a,c) &= \mathbb{P}(X \leq a,\, Y \leq d) - \mathbb{P}(X \leq a,\, Y \leq c) \\ &= \mathbb{P}(X \leq a,\, c < Y \leq d) \end{aligned}$$

   Subtracting the second equation above from the first,

$$\begin{aligned} F_{X,Y}(b,d) - F_{X,Y}(b,c) - F_{X,Y}(a,d) + F_{X,Y}(a,c) &= \mathbb{P}(X \leq b,\, c < Y \leq d) - \mathbb{P}(X \leq a,\, c < Y \leq d) \\ &= \mathbb{P}(a < X \leq b,\, c < Y \leq d) \end{aligned}$$

   as required.

4. Random variables X and Y have joint probability mass function given by the following:

	y=0	1	2	3
x=0	0	3/56	6/56	1/56
1	3/56	18/56	9/56	0
2	6/56	9/56	0	0
3	1/56	0	0	0

   (a)

$$\begin{aligned} \mathbb{P}(X < Y) &= \mathbb{P}(X=0,Y=1) + \mathbb{P}(X=0,Y=2) + \mathbb{P}(X=0,Y=3) \\ &\quad + \mathbb{P}(X=1,Y=2) + \mathbb{P}(X=1,Y=3) + \mathbb{P}(X=2,Y=3) \\ &= {}^{3}\!/_{56} + {}^{6}\!/_{56} + {}^{1}\!/_{56} + {}^{9}\!/_{56} + 0 + 0 \\ &= {}^{19}\!/_{56} \approx 0.3393 \end{aligned}$$

   (b) Marginal pmf for X:

$$\begin{aligned} p_X(0) = \mathbb{P}(X=0) &= \mathbb{P}(X=0,Y=0) + \mathbb{P}(X=0,Y=1) + \mathbb{P}(X=0,Y=2) \\ &\quad + \mathbb{P}(X=0,Y=3) \\ &= 0 + {}^{3}\!/_{56} + {}^{6}\!/_{56} + {}^{1}\!/_{56} = {}^{10}\!/_{56} \\ p_X(1) = \mathbb{P}(X=1) &= \mathbb{P}(X=1,Y=0) + \mathbb{P}(X=1,Y=1) + \mathbb{P}(X=1,Y=2) \\ &\quad + \mathbb{P}(X=1,Y=3) \\ &= {}^{3}\!/_{56} + {}^{18}\!/_{56} + {}^{9}\!/_{56} + 0 = {}^{30}\!/_{56} \\ p_X(2) = \mathbb{P}(X=2) &= \mathbb{P}(X=2,Y=0) + \mathbb{P}(X=2,Y=1) + \mathbb{P}(X=2,Y=2) \\ &\quad + \mathbb{P}(X=2,Y=3) \\ &= {}^{6}\!/_{56} + {}^{9}\!/_{56} + 0 + 0 = {}^{15}\!/_{56} \\ p_X(3) = \mathbb{P}(X=3) &= \mathbb{P}(X=3,Y=0) + \mathbb{P}(X=3,Y=1) + \mathbb{P}(X=3,Y=2) \\ &\quad + \mathbb{P}(X=3,Y=3) \\ &= {}^{1}\!/_{56} + 0 + 0 + 0 = {}^{1}\!/_{56} \end{aligned}$$

   So

   i. $\mathbb{P}(X \geq 2) = p_X(2) + p_X(3) = {}^{15}\!/_{56} + {}^{1}\!/_{56} = {}^{16}\!/_{56} = {}^{2}\!/_{7} \approx 0.2857$

   ii. $\mathbb{E}(X) = 0 \times p_X(0) + 1 \times p_X(1) + 2 \times p_X(2) + 3 \times p_X(3) = 0 \times {}^{10}\!/_{56} + 1 \times {}^{30}\!/_{56} + 2 \times {}^{15}\!/_{56} + 3 \times {}^{1}\!/_{56} = {}^{63}\!/_{56} = {}^{9}\!/_{8} = 1.125$

(c) Marginal pmf for Y:

$$\begin{aligned}
p_Y(0) = \mathbb{P}(Y=0) &= \mathbb{P}(X=0,Y=0) + \mathbb{P}(X=1,Y=0) + \mathbb{P}(X=2,Y=0) \\
&\quad + \mathbb{P}(X=3,Y=0) \\
&= 0 + {}^{3}\!/_{56} + {}^{6}\!/_{56} + {}^{1}\!/_{56} = {}^{10}\!/_{56} \\
p_Y(1) = \mathbb{P}(Y=1) &= \mathbb{P}(X=0,Y=1) + \mathbb{P}(X=1,Y=1) + \mathbb{P}(X=2,Y=1) \\
&\quad + \mathbb{P}(X=3,Y=1) \\
&= {}^{3}\!/_{56} + {}^{18}\!/_{56} + {}^{9}\!/_{56} + 0 = {}^{30}\!/_{56} \\
p_Y(2) = \mathbb{P}(Y=2) &= \mathbb{P}(X=0,Y=2) + \mathbb{P}(X=1,Y=2) + \mathbb{P}(X=2,Y=2) \\
&\quad + \mathbb{P}(X=3,Y=2) \\
&= {}^{6}\!/_{56} + {}^{9}\!/_{56} + 0 + 0 = {}^{15}\!/_{56} \\
p_Y(3) = \mathbb{P}(Y=3) &= \mathbb{P}(X=0,Y=3) + \mathbb{P}(X=1,Y=3) + \mathbb{P}(X=2,Y=3) \\
&\quad + \mathbb{P}(X=3,Y=3) \\
&= {}^{1}\!/_{56} + 0 + 0 + 0 = {}^{1}\!/_{56}
\end{aligned}$$

So

i. $\mathbb{P}(Y \leq 1) = p_Y(0) + p_Y(1) = {}^{10}\!/_{56} + {}^{30}\!/_{56} = {}^{40}\!/_{56} = {}^{5}\!/_{7} \approx 0.7143$

ii. $\mathbb{E}(Y) = 0 \times p_Y(0) + 1 \times p_Y(1) + 2 \times p_Y(2) + 3 \times p_Y(3) = 0 \times {}^{10}\!/_{56} + 1 \times {}^{30}\!/_{56} + 2 \times {}^{15}\!/_{56} + 3 \times {}^{1}\!/_{56} = {}^{63}\!/_{56} = {}^{9}\!/_{8} = 1.125$

(d) Pmf for $Z = X - Y$: first notice that $\mathrm{Im}Z = \{-3,-2,-1,0,1,2,3\}$, and then

$$\begin{aligned}
p_Z(-3) &= \mathbb{P}(Z=-3) = \mathbb{P}(X-Y=-3) = \mathbb{P}(X=0,Y=3) = {}^{1}\!/_{56} \\
p_Z(-2) &= \mathbb{P}(Z=-2) = \mathbb{P}(X-Y=-2) \\
&= \mathbb{P}(X=0,Y=2) + \mathbb{P}(X=1,Y=3) = {}^{6}\!/_{56} + 0 = {}^{6}\!/_{56} \\
p_Z(-1) &= \mathbb{P}(Z=-1) = \mathbb{P}(X-Y=-1) \\
&= \mathbb{P}(X=0,Y=1) + \mathbb{P}(X=1,Y=2) + \mathbb{P}(X=2,Y=3) = {}^{3}\!/_{56} + {}^{9}\!/_{56} + 0 = {}^{12}\!/_{56} \\
p_Z(0) &= \mathbb{P}(Z=0) = \mathbb{P}(X-Y=0) \\
&= \mathbb{P}(X=0,Y=0) + \mathbb{P}(X=1,Y=1) + \mathbb{P}(X=2,Y=2) + \mathbb{P}(X=3,Y=3) \\
&= 0 + {}^{18}\!/_{56} + 0 + 0 = {}^{18}\!/_{56} \\
p_Z(1) &= \mathbb{P}(Z=1) = \mathbb{P}(X-Y=1) \\
&= \mathbb{P}(X=1,Y=0) + \mathbb{P}(X=2,Y=1) + \mathbb{P}(X=3,Y=2) = {}^{3}\!/_{56} + {}^{9}\!/_{56} + 0 = {}^{12}\!/_{56} \\
p_Z(2) &= \mathbb{P}(Z=2) = \mathbb{P}(X-Y=2) \\
&= \mathbb{P}(X=2,Y=0) + \mathbb{P}(X=3,Y=1) = {}^{6}\!/_{56} + 0 = {}^{6}\!/_{56} \\
p_Z(3) &= \mathbb{P}(Z=3) = \mathbb{P}(X-Y=3) = \mathbb{P}(X=3,Y=0) = {}^{1}\!/_{56}
\end{aligned}$$

So

i. $\mathbb{P}(-1 \leq Z \leq 1) = p_Z(-1) + p_Z(0) + p_Z(1) = {}^{12}\!/_{56} + {}^{18}\!/_{56} + {}^{12}\!/_{56} = {}^{42}\!/_{56} = {}^{3}\!/_{4} = 0.75$

ii. The pmf $p_Z(z)$ is symmetric about 0, so $\mathbb{E}(Z) = 0$.

iii.

$$\mathbb{E}(Z^2) = (-3)^2\frac{1}{56} + (-2)^2\frac{6}{56} + (-1)^2\frac{12}{56} + 0^2\frac{18}{56} + 1^2\frac{12}{56} + 2^2\frac{6}{56} + 3^2\frac{1}{56} = \frac{45}{28}.$$

Therefore

$$\sigma_Z = \sqrt{\mathbb{E}(Z^2) - \mathbb{E}(Z)^2} = \sqrt{\frac{45}{28} - 0^2} \approx 1.2677.$$

(e) Conditional pmf of X given $Y = 0$ is

$$\begin{aligned} p_{X|Y}(0 \mid 0) &= \frac{p_{X,Y}(0,0)}{p_Y(0)} = \frac{0}{10/56} = 0 \\ p_{X|Y}(1 \mid 0) &= \frac{p_{X,Y}(1,0)}{p_Y(0)} = \frac{3/56}{10/56} = 3/10 \\ p_{X|Y}(2 \mid 0) &= \frac{p_{X,Y}(2,0)}{p_Y(0)} = \frac{6/56}{10/56} = 6/10 \\ p_{X|Y}(3 \mid 0) &= \frac{p_{X,Y}(3,0)}{p_Y(0)} = \frac{1/56}{10/56} = 1/10 \end{aligned}$$

So

i. $\mathbb{P}(X \geq 2 \mid Y = 0) = p_{X|Y}(2 \mid 0) + p_{X|Y}(3 \mid 0) = 6/10 + 1/10 = 7/10 = 0.7$

ii. $\mathbb{E}(X \mid Y = 0) = 1 \times p_{X|Y}(1 \mid 0) + 2 \times p_{X|Y}(2 \mid 0) + 3 \times p_{X|Y}(3 \mid 0) = 1 \times 3/10 + 2 \times 6/10 + 3 \times 1/10 = 18/10 = 1.8$

(f) Conditional pmf of Y given $X = 1$ is

$$\begin{aligned} p_{Y|X}(0 \mid 1) &= \frac{p_{X,Y}(1,0)}{p_X(1)} = \frac{3/56}{30/56} = 3/30 = 1/10 \\ p_{Y|X}(1 \mid 1) &= \frac{p_{X,Y}(1,1)}{p_X(1)} = \frac{18/56}{30/56} = 18/30 = 6/10 \\ p_{Y|X}(2 \mid 1) &= \frac{p_{X,Y}(1,2)}{p_X(1)} = \frac{9/56}{30/56} = 9/30 = 3/10 \\ p_{Y|X}(3 \mid 1) &= \frac{p_{X,Y}(1,3)}{p_X(1)} = \frac{0}{30/56} = 0 \end{aligned}$$

So

i. $\mathbb{P}(Y \leq 1 \mid X = 1) = p_{Y|X}(0 \mid 1) + p_{Y|X}(1 \mid 1) = 1/10 + 6/10 = 7/10 = 0.7$

ii. $\mathbb{E}(Y \mid X = 1) = 0 \times p_{Y|X}(0 \mid 1) + 1 \times p_{Y|X}(1 \mid 1) + 2 \times p_{Y|X}(2 \mid 1) = 0 \times 1/10 + 1 \times 6/10 + 2 \times 3/10 = 12/10 = 1.2$

(g) The random variables X, Y are <u>not</u> independent, because (for example)

$$\begin{aligned} \mathbb{P}(X = 0, Y = 0) &= 0 \\ \mathbb{P}(X = 0)\mathbb{P}(Y = 0) &= 10/56 \times 10/56 \neq 0 \end{aligned}$$

so $\mathbb{P}(X = 0, Y = 0) \neq \mathbb{P}(X = 0)\mathbb{P}(Y = 0)$, from which it follows that X, Y are not independent.

5. (a) Suppose that $X \sim \text{Exp}(\lambda)$ where $\lambda > 0$.
Then for any $x \geq 0$,

$$\mathbb{P}(X > x) = 1 - \mathbb{P}(X \leq x) = 1 - F_X(x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$$

So now for any $s, t > 0$,

$$\begin{aligned} \mathbb{P}(X > s + t \mid X > t) &= \frac{\mathbb{P}(\{X > s + t\} \cap \{X > t\})}{\mathbb{P}(X > t)} \\ &= \frac{\mathbb{P}(X > s + t)}{\mathbb{P}(X > t)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda t}} \\ &= e^{-\lambda s} \\ &= \mathbb{P}(X > s) \end{aligned}$$

as required.

(b) Since T is exponentially distributed with mean 2 (hours), we have $T \sim \text{Exp}(1/2)$

i. Using result obtained towards the start of part (a), $\mathbb{P}(T > 2) = e^{-(1/2)2} = e^{-1} \approx 0.3679$

ii. Using the memoryless property from part (a), $\mathbb{P}(T > 6 \mid T > 4) = \mathbb{P}(T > 2) = e^{-1} \approx 0.3679$

Exam Style Question

(a) i) Note that $T \leq t$ if and only if $X_1 \leq t$ and $X_2 \leq t$ so

$$F_T(t) = \mathbb{P}(T < t) = \mathbb{P}(X_1 < t, X_2 < t)$$

Since X_1 and X_2 are independent

$$F_T(t) = \mathbb{P}(T \leq t) = \mathbb{P}(X_1 \leq t, X_2 \leq t) = \mathbb{P}(X_1 \leq t)\mathbb{P}(X_2 \leq t).$$

Since X_1 and X_2 have the same distribution we will denote their cumulative distribution function as F_X then

$$F_T(t) = \mathbb{P}(X_1 \leq t)\mathbb{P}(X_2 \leq t) = F_X(t)^2.$$

Since $X_1, X_2 \sim \text{Exp}(\lambda^{-1})$ we have for $x \geq 0$

$$F_X(x) = 1 - e^{-\lambda^{-1}x}$$

and $F_X(x) = 0$ for $x < 0$.
Hence for $t \geq 0$

$$F_T(t) = (1 - e^{-\lambda^{-1}t})^2$$

and $F_T(t) = 0$ for $t < 0$. [4 marks]

We can find the probability density function of T by differentiating F_T. By the chain rule for $t > 0$

$$f_T(t) = F_T'(t) = \frac{2}{\lambda}e^{-\lambda^{-1}t}(1 - e^{-\lambda^{-1}t}).$$

For $t < 0$ $f_T(t) = 0$.

[2 marks]

ii) The expected value of T can be calculated from the probability density function as follows:

$$\begin{aligned}
\mathbb{E}[T] &= \int_0^\infty t f_T(t)\,\mathrm{d}t \\
&= \int_0^\infty t \cdot \frac{2}{\lambda}e^{-\lambda^{-1}t}(1 - e^{-\lambda^{-1}t})\,\mathrm{d}t \\
&= \frac{2}{\lambda}\int_0^\infty te^{-\lambda^{-1}t}\,\mathrm{d}t - \frac{2}{\lambda}\int_0^\infty te^{-2\lambda^{-1}t}\,\mathrm{d}t
\end{aligned}$$

Let $X_\lambda \sim \text{Exp}(\lambda^{-1})$ and $X_{2\lambda} \sim \text{Exp}(2\lambda^{-1})$ then we have

$$\begin{aligned}
\int_0^\infty t\lambda^{-1}e^{-\lambda^{-1}t}\,\mathrm{d}t &= \mathbb{E}[X_\lambda] = \lambda, \\
\int_0^\infty t \cdot \frac{2}{\lambda}e^{-2\lambda^{-1}t}\,\mathrm{d}t &= \mathbb{E}[X_{2\lambda}] = \frac{1}{2}\lambda.
\end{aligned}$$

Therefore

$$\mathbb{E}[T] = 2\mathbb{E}[X_\lambda] - \mathbb{E}[X_{2\lambda}] = 2\lambda - \frac{1}{2}\lambda = \frac{3}{2}\lambda.$$

[5 marks]

(b) Let X_1 be the lifetime of the first bulb and X_2 be the lifetime of the second bulb. We will assume that $X_1, X_2 \sim \text{Exp}(1)$ are independent. Let $T = \max(X_1, X_2)$ then the lamp is still functioning after 2 years corresponds to the event that $T > 2$.

Using part (a) with $\lambda = 1$

$$\mathbb{P}(T > 2) = 1 - F_T(2) = 1 - (1 - e^{-2})^2 = 2e^{-2} - e^{-4}.$$

[2 marks]